

MASTER'S THESIS

Thesis submitted in fulfillment of the requirements for the degree of Master of Science in Engineering at the University of Applied Sciences Technikum Wien - Degree Program Artificial Intelligence Engineering

Titel der Arbeit

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Wien, July 25, 2026

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Schlagworte: Schlagwort1, Schlagwort2, Schlagwort3, Schlagwort4

Abstract

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Keywords: Keyword1, Keyword2, Keyword3, Keyword4

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1 Erste Überschrift der Ebene 1 (chapter)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{i=n} x_i = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

$$\int_0^{\infty} e^{-\alpha x^2} dx = \frac{1}{2} \sqrt{\int_{-\infty}^{\infty} e^{-\alpha x^2} dx \int_{-\infty}^{\infty} e^{-\alpha y^2} dy} = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

$$\sum_{k=0}^{\infty} a_0 q^k = \lim_{n \rightarrow \infty} \sum_{k=0}^n a_0 q^k = \lim_{n \rightarrow \infty} a_0 \frac{1 - q^{n+1}}{1 - q} = \frac{a_0}{1 - q}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font,

how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2}$$

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

1.1 Erste Überschrift Tiefe 2 (section)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

1.2 Erste Überschrift Tiefe 2 (section)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font,

how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

1.2.1 Erste Überschrift Tiefe 3 (subsection)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Erste Überschrift Tiefe 4 (subsubsection)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

1.2.2 Erste Überschrift Tiefe 3 (subsection)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

2 Zweite Überschrift der Tiefe 1 (chapter)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there

no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

2.1 Zweite Überschrift Tiefe 2 (section)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

2.2 Zweite Überschrift Tiefe 2 (section)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

2.2.1 Zweite Überschrift Tiefe 3 (subsection)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

2.2.2 Dritte Überschrift Tiefe 3 (subsection)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Zweite Überschrift Tiefe 4 (subsubsection)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Querverweise werden in $\text{\LaTeX}$ automatisch erzeugt und verwaltet, damit sie leicht aktualisiert werden können. Hier wird zum Beispiel auf Abbildung 1 verwiesen.

Einstein Albert **2008**

Figure 1: Beispiel für die Beschriftung eines Buchrückens.

Einstein Albert **2008**

Figure 2: 2. Beispiel für die Beschriftung eines Buchrückens.

Und hier ist ein Verweis auf Tabelle 1. Das gezeigte Tabellenformat ist nur ein Beispiel. Tabellen können individuell gestaltet werden.

Table 1: Semesterplan der Lehrveranstaltung „Angewandte Mathematik“.

Datum	Thema	Raum
20.08.2008	Graphentheorie	HS 3.13
01.10.2008	Biomathematik	HS 1.05

Hier wird auf die Formel 1 verwiesen.

Table 2: 2. Semesterplan der Lehrveranstaltung „Angewandte Mathematik“.

Datum	Thema	Raum
20.08.2008	Graphentheorie	HS 3.13
01.10.2008	Biomathematik	HS 1.05

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q} \quad (1)$$

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q} \quad (2)$$

Literaturverweise sollten automatisch verwaltet werden, vor allem, wenn es viele Quellenverweise gibt. Beispiele sind [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14]. Das verwendete Zitierformat (bzw. das Format des Literaturverzeichnisses) ist entsprechend der Vorgaben der Studiengänge zu wählen. Es wird dringend empfohlen, Biber oder BibTeX zu verwenden (wie in diesen Beispielen).

3 Dritte Überschrift der Tiefe 1 (chapter)

Einstein Albert 2008

Figure 3: 3. Beispiel für die Beschriftung eines Buchrückens.

Einstein Albert 2008

Figure 4: 4. Beispiel für die Beschriftung eines Buchrückens.

Table 3: 3. Semesterplan der Lehrveranstaltung „Angewandte Mathematik“.

Datum	Thema	Raum
20.08.2008	Graphentheorie	HS 3.13
01.10.2008	Biomathematik	HS 1.05

Table 4: 4. Semesterplan der Lehrveranstaltung „Angewandte Mathematik“.

Datum	Thema	Raum
20.08.2008	Graphentheorie	HS 3.13
01.10.2008	Biomathematik	HS 1.05

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q} \quad (3)$$

$$x = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q} \quad (4)$$

The following commands are provided to provide quality of life improvements

- `\citepic` the usage of the command is depicted in figure 5.
- `\citefig` the usage of the command is depicted in figure 6.
- `\citefigm` the usage of the command is depicted in figure 7.

Einstein Albert 2008

Figure 5: Usage of the command `\citepic`: (Source: Anglia Ruskin University Library [15])

Einstein Albert 2008

Figure 6: usage of the command `\citefig`: (Source: Anglia Ruskin University Library [15], p. 30)

Einstein Albert 2008

Figure 7: usage of the command `\citefigm`: (Source: taken with modification from Anglia Ruskin University Library [15], p. 30)

Bibliography

- [1] H. Kopka, *LaTeX, Band 1: Einführung*, 3rd ed. München: Pearson Studium, 2005.
- [2] H. Kopka, *LaTeX, Band 1: Einführung*, 3rd ed. München: Pearson Studium, 2005. Accessed: Jul. 6, 2011. [Online]. Available: <http://www.pearson-studium.de>
- [3] M. Goossens, F. Mittelbach, and A. Samarin, *Der LaTeX Begleiter*. Bonn: Addison-Wesley Deutschland, 2002.
- [4] S. Teschl, K. M. Göschka, and G. Essl, *Leitfaden zur Verfassung einer Bachelorarbeit oder Master Thesis*, FH Technikum Wien, 2014. Accessed: Aug. 4, 2014. [Online]. Available: www.technikum-wien.at
- [5] M. Humenberger, D. Hartermann, and W. Kubinger, "Evaluation of stereo matching systems for real world applications using structured light for ground truth estimation," in *Proceedings of the Tenth IAPR Conference on Machine Vision Applications (MVA2007)*, Tokyo, Japan: MVA Conference Committee, 2007, pp. 433–436.
- [6] M. Humenberger, C. Zinner, M. Weber, W. Kubinger, and M. Vincze, "A fast stereo matching algorithm suitable for embedded real-time systems," *Computer Vision and Image Understanding*, vol. 114, no. 11, pp. 1180–1202, 2010.
- [7] C. Zinner, W. Kubinger, and R. Isaacs, "Pfelib: A performance primitives library for embedded vision," *EURASIP Journal on Embedded Systems*, vol. 2007, pp. 1–14, 2007. Accessed: Jul. 6, 2011. [Online]. Available: <http://downloads.hindawi.com/journals/es/2007/049051.pdf>
- [8] H. Hemetsberger, *Ait stereo sensor im einsatz während der darpa urban challenge 2007*, AIT Austrian Institute of Technology, 2007.
- [9] Siemens Automation Technology. "Simatic," Accessed: Jul. 6, 2011. [Online]. Available: <http://www.automation.siemens.com/mcms/topics/de/simatic/Seiten/Default.aspx>
- [10] Siemens Automation Technology. "Simatic," Accessed: Oct. 17, 2014. [Online]. Available: <http://www.automation.siemens.com/mcms/topics/de/simatic/Seiten/Default.aspx>
- [11] International Standards Office, *Iso 690 – information and documentation: Bibliographical references: Electronic documents*, Genf: International Standards Office, 1998.
- [12] Atmel Corporation, *Atmel atmega16 – 8-bit microcontroller with 16k bytes in-system programmable flash*, San Jose, United States: Atmel Corporation, 2011. Accessed: Jul. 6, 2011. [Online]. Available: http://www.atmel.com/dyn/resources/prod%5C_documents/doc2466.pdf

- [13] M. Humenberger, "Real-time stereo matching for embedded systems in robotic applications," Dissertation, Technische Universität Wien, Fakultät für Elektrotechnik und Informationstechnik, Wien, 2011.
- [14] J. Pohn, "Condition monitoring systeme für die zustandorientierte instandhaltung von windkraftanlagen," Diplomarbeit, FH Technikum Wien, Masterstudiengang Innovations- und Technologiemanagement, Wien, 2010.
- [15] Anglia Ruskin University Library, *Guide to the harvard style of referencing*, Cambridge, UK: Anglia Ruskin University, 2010. Accessed: Jul. 6, 2011. [Online]. Available: <http://libweb.anglia.ac.uk>

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Documentation table of AI-based tools

AI-Based Tool	Intended Use	Prompt, Source, Page, Paragraph. . .
DeepL Translate	Translation of an article in English	Source (XXX), Chapter X on page X-X
Chat GPT 4.0	Grammar and Spelling	"Please list issues with spelling and grammar in the following text: ...", Entire document

List of Abbreviations

ABC Alphabet

WWW world wide web

ROFL Rolling on floor laughing

A Anhang A

B Anhang B